
Music2Scene: Audio-Driven Visual Generation via Learned Prompt Embeddings

September 30, 2026

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Abstract

While recent advancements in audio-driven image synthesis have successfully mapped complex acoustic scenes to high-fidelity static frames, capturing the rhythmic and narrative pulse of a musical composition remains a challenge.

In this paper, we propose Music2Scene, a framework designed to generate a coherent and synchronized sequence of discrete visual snapshots that evolve alongside the input audio.

By implementing a deep Audio-to-Prompt Adapter with residual connections, our method translates 8-second audio slices into high-dimensional latent tokens compatible with the CLIP space. By treating the audio input as a dynamic guide for a multi-step visual narrative, we move beyond the static snapshot focus of previous works to capture the temporal vibrancy of the soundscape.

tent prompt space of diffusion models. Our system prioritizes a high-fidelity visual collage where each image corresponds to a temporal slice of the audio. By learning a transformation that produces CLIP-compatible prompt embeddings, we enable integration between auditory features and pre-trained models like OpenJourney-v4.

Our main contribution is a learned audio-to-prompt adapter that aligns CLAP audio embeddings with CLIP text embeddings, allowing audio to act as a first-class conditioning signal for image generation.

Unlike standard methods that treat audio as a stationary conditioning signal, our implementation utilizes a real-time slicing and inference pipeline to treat audio as a dynamic guide for a multi-step visual narrative. By dividing the input into discrete segments, we produce a visual output that mirrors the intensity and content changes of the sound. This approach captures the punctuated pulse of the music: it generates a sequence that maintains stylistic consistency while reflecting the shifting dynamics of the soundscape and produces a rhythmic montage where each frame is a direct visual manifestation of the audio’s intensity and texture. The aim is to be able to use it live during a concert.

1. Introduction

Recent breakthroughs in generative models have shifted focus from simple text-to-image mapping toward leveraging the rich, high-frequency information found in audio. As we will see in the next paragraph, models like AudioToken and SonicDiffusion have demonstrated that lightweight adapters can effectively project auditory features into the latent space of pre-trained diffusion backbones without the need for exhaustive retraining. However, these state-of-the-art methods are primarily optimized for generating single, high-quality snapshots of an acoustic scene.

In this work, we propose **Music2Scene**, a system that generates a rhythmic sequence of images conditioned on audio signals by directly mapping audio embeddings into the la-

2. Related Work

Recent advancements in generative modeling have been dominated by diffusion-based models, such as Stable Diffusion, which leverage CLIP-based text embeddings to achieve results in high-quality image synthesis. These models rely on strong alignment between text and visual representations; however, while they excel at text-to-image tasks, natural language often fails to capture the high-frequency details of acoustic signals. Relying on manual text annotations for audio is labor-intensive and frequently overlooks the subtle textures that define a soundscape.

Early baselines in the audio-to-image space include Wav2Clip, which utilizes a CLIP-like architecture for audio-image pairs via VQ-GAN, and ImageBind, which aligns multiple modalities into a single representation space for zero-shot generation. While foundational, these approaches often rely on intermediate textual representa-

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tions or struggle to capture the fine-grained semantic details found in complex acoustic scenes.

To address these limitations, the AudioToken framework introduced a lightweight adaptation layer that encodes audio signals into a dedicated representation. This representation functions as a "new word" token within the text-conditioned latent space. By training only a small Embedder network while keeping the pre-trained generative backbone frozen, this method avoids the computational cost of full model retraining while producing images that are more semantically accurate than text-only baselines.

Building further upon these concepts, SonicDiffusion proposed a more sophisticated framework that utilizes a multi-token representation ($K=77$ tokens) to model the relationship between audio and image features. Unlike previous single-token methods, which may lack the capacity to distinguish between complex audio inputs, SonicDiffusion employs gated cross-attention layers to integrate audio features directly into the diffusion process. This enables not only high-fidelity generation but also precise audio-driven image editing.

3. Method

Our proposed framework, Music2Scene, is designed to bridge the gap between auditory features and visual synthesis by aligning the embedding spaces of two models: CLAP (Audio-Text) and CLIP (Image-Text).

3.1. Baselines

As a starting point, we consider the architecture proposed in AudioToken, which utilizes a lightweight, single-layer linear projection to map audio features into a single 1×768 token representation. We implemented a preliminary version of this approach using a basic sequential network consisting of two linear layers and a GELU activation function to map the 512-dimensional CLAP output to the required CLIP latent dimension. While this naive solution successfully triggers the diffusion backbone, it often fails to capture the intricate stylistic nuances of complex musical compositions, leading to visual results that lack diversity and textural detail.

3.2. Contribution

Our primary contribution is a robust, multi-layer **Audio-to-Prompt Adapter** and a synchronized temporal pipeline designed for rhythmic visual narratives.

3.2.1. ADAPTER ARCHITECTURE

Unlike the baseline, our final adapter utilizes a deep feed-forward network consisting of $N = 3$ layers with Resid-

ual Connections and LayerNorm to stabilize training. The model f_θ learns a mapping from the CLAP embedding space $\mathcal{A} \in \mathbb{R}^{512}$ to a multi-token prompt space $\mathcal{P} \in \mathbb{R}^{K \times 768}$ (`adapter.py`):

$$f_\theta : \mathbb{R}^{512} \rightarrow \mathbb{R}^{K \times 768}$$

$$\mathbf{P}_{audio} = f_\theta(\mathbf{E}_{clap})$$

where K is the number of tokens (optimized at $K = 4$ for our experiments). The inclusion of residual projections allows the model to preserve the original auditory energy while the deeper layers extract complex semantic features.

3.2.2. TRAINING OBJECTIVES

To ensure the generated images are both semantically relevant and visually diverse, the adapter is optimized using a multi-objective composite loss function (`train_adapter_robust.py`)

$$\mathcal{L}_{tot} = \mathcal{L}_{contr} + \lambda_{div}\mathcal{L}_{div} + \lambda_{var}\mathcal{L}_{var}$$

where the hyper-parameters λ_{div} and λ_{var} balance the following components:

- **Contrastive Loss ($\mathcal{L}_{contr}$):** an InfoNCE-based objective (Information Noise-Contrastive Estimation) employed to achieve semantic alignment between modalities. It maximizes the cosine similarity between matching audio-text pairs while simultaneously minimizing the similarity between mismatched negative pairs within the training batch;
- **Diversity Loss ($\mathcal{L}_{div}$):** designed to prevent the generation of repetitive imagery. This objective penalizes high pairwise cosine similarity between different samples within a single batch, ensuring the generator produces distinct visual outputs for varying auditory inputs;
- **Variance Loss ($\mathcal{L}_{var}$):** aimed at preventing representation collapse by encouraging a higher standard deviation across the embedding dimensions. This ensures that the predicted audio tokens occupy a broad and informative region of the latent space rather than collapsing into a narrow manifold.

3.2.3. TEMPORAL SLICING AND REAL-TIME PIPELINE

To capture the temporal evolution shown in our results, we implemented a real-time slicing mechanism. The input audio $\mathcal{S}$ is divided into discrete windows of duration $\Delta = 8.0s$:

$$\mathcal{S} = \{chunk_1, chunk_2, \dots, chunk_n\}$$

Each chunk is processed independently through the inference pipeline, where the audio tokens are concatenated with a static base prompt to maintain thematic consistency while allowing the audio to modulate style and intensity (`infl.py`). The concatenated embedding is used as conditioning input for a pretrained Stable Diffusion model to generate images. Finally, a high-fidelity VAE (`sd-vae-ft-mse`) is employed to reconstruct the latent representations into sharp, 512×512 visual frames.

4. Experiments and Results

We evaluated Music2Scene by monitoring the alignment between auditory and textual latent spaces during training and performing a qualitative analysis of the resulting synchronized video.

4.1. Quantitative Evaluation

The effectiveness of the Audio-to-Prompt adapter is measured by its ability to project CLAP embeddings into the CLIP latent space while maintaining sufficient feature discriminability.

To quantify this behavior, we analyzed the statistical distribution of the embeddings before (`analyze_audio_diversity.py`) and after (`diagnostic_adapter_diversity.py`) the training using two primary metrics:

- **Pairwise Cosine Similarity:** calculated between embedding pairs to assess latent overlap. High average similarity indicates a crowded latent space and potential mode collapse, whereas lower values suggest successful differentiation between auditory concepts;
- **Feature Variance:** calculated as the mean variance across all embedding dimensions. This serves as a proxy for the expressiveness of the features; low variance implies a collapsed representation, while higher variance reflects a richer, more detailed mapping.

4.1.1. ACOUSTIC BASELINE VS. ADAPTER OUTPUT

We first analyzed the raw CLAP embeddings from the MusicCaps classical music subset ($N=5306$). The baseline results showed an intrinsic acoustic variance of 0.0326 and a cosine similarity of 0.4454 ± 0.1755 . These relatively high similarity and low variance are expected given the stylistic homogeneity of the classical genre, providing a challenging starting point for distinct image generation.

After the training of the adapter (final training loss: 0.3426), we re-evaluated the diversity of the generated

prompt tokens. The results demonstrate significant feature amplification:

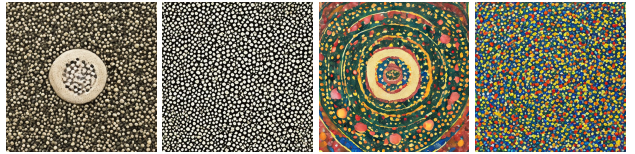
- **Variance Expansion:** the mean variance increased from 0.0326 to 0.7074, indicating that the adapter successfully expanded the compressed acoustic features into a broader region of the CLIP latent space;
- **Latent Decorrelation:** the average cosine similarity dropped to 0.004 ± 0.174 , suggesting that the adapter effectively decorrelated different audio samples.

These results confirm that even within a single musical genre, our framework successfully translates subtle acoustic variations into highly distinct latent prompts. This ensures that the downstream diffusion model receives a unique visual instruction for every audio slice, directly contributing to the rhythmic diversity observed in our generated sequences.

4.2. Qualitative Results

To evaluate the temporal responsiveness of the framework, we analyzed the generated sequence for a heterogeneous composite file, `input.wav`, which merged different acoustic sources including white noise, full vocal compositions, and isolated instrumental recordings.

The resulting sequence aligned precisely with the input’s acoustic properties: white noise segments yielded relatively static visual representations, while transitions between distinct auditory sections, such as shifts from noise to music or from vocals to instrumentation, triggered abrupt visual changes. Crucially, these transitions maintained strong thematic consistency due to the concatenation of the learned audio tokens with the fixed base prompt. In contrast, images generated from sequential slices of the same musical piece exhibited greater stylistic continuity and smoother semantic transitions.



These observations validate that the latent prompt embeddings generated by our adapter successfully capture both the spectral stability and the semantic evolution of the audio signal.

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Appendix

A. Evolution of the Architecture

This appendix provides a detailed account of the iterative research process that led to the final implementation of Music2Scene. We document the failed architectural baselines, parameter searches, and the reasons behind our final design choices.

Adapter Architecture

Our adapter architecture underwent a phased evolution to find the optimal balance between representational capacity and training stability. Our initial attempt utilized a 1-layer baseline consisting of a simple sequential network with only a single GELU activation positioned between two linear layers. This configuration proved insufficient for capturing the intricate spectral nuances of complex audio, resulting in visual outputs that lacked textural detail. We subsequently scaled to an 8-layer deep network; however, this configuration introduced unnecessary complexity.

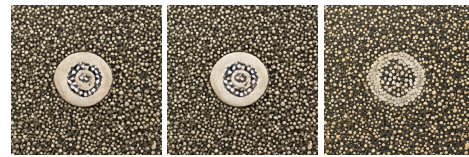
Consequently, we settled on a 3-layer residual adapter as our final architecture. In this final version, the model achieves a more robust feature extraction through a series of blocks, each characterized by a linear transformation followed by a GELU non-linear activation function and LayerNorm to stabilize the latent distribution. This design represents a significant evolution from the original version: by

integrating these non-linearities within a residual framework, the model is able to preserve the original auditory energy while effectively mapping audio features into the $K \times 768$ CLIP latent space.

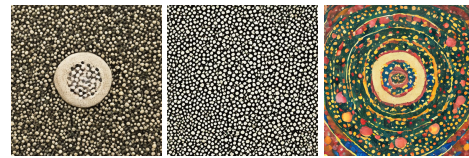
Token Optimization (K)

Another critical parameter in our architecture is K , the number of tokens, which defines the “visual vocabulary” size the adapter utilizes to communicate auditory features to the Stable Diffusion generator. We observed that increasing K enhances the model’s sensitivity to acoustic nuances, while decreasing K ensures the generator remains faithful to the text prompt.

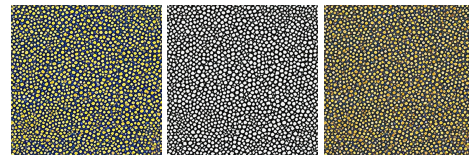
Following the *AudioToken* methodology, we initially evaluated a single-token representation ($K = 1$), which proved to be an information bottleneck, meaning not enough data could get through; the resulting generic snapshots largely ignored the complex spectral textures of the input music. In contrast, we experimented with the high-capacity approach proposed in *SonicDiffusion*, utilizing $K = 77$ tokens (the maximum capacity of the CLIP text encoder). However, this led to chaotic visual artifacts, as the model started interpreting every tiny bit of audio data as a visual instruction. After iterative empirical testing, we settled on $K = 4$. This value provides sufficient representational capacity for the adapter to describe complex choral and instrumental textures while maintaining the thematic integrity of the user-defined prompt.



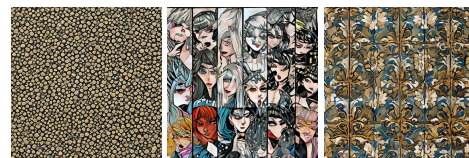
(a) $K = 1$



(b) $K = 4$



(c) $K = 10$



(d) $K = 77$

Error Function

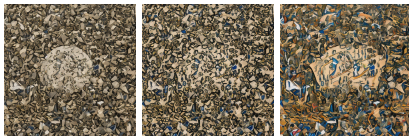
During the initial training phase of the project, we implemented a Mean Squared Error (MSE) loss function. This approach aimed to minimize the point-to-point Euclidean distance between the adapter’s predicted tokens $\mathbf{P}_{\text{audio}}$ and the target CLIP text embeddings $\mathbf{E}_{\text{text}}$, while already incorporating a diversity penalty to encourage variation:

$$\mathcal{L}_{\text{tot}} = \frac{1}{N} \sum_{i=1}^N |\mathbf{P}_{\text{audio}}(i) - \mathbf{E}_{\text{text}}(i)|^2 + \lambda_{\text{div}} \mathcal{L}_{\text{div}}.$$

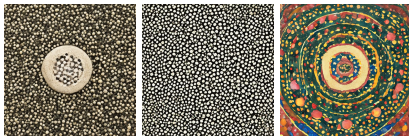
Experimental monitoring revealed a Validation Cosine Similarity approaching 1.0. This result indicated that the MSE loss was too restrictive for the high-dimensional CLIP latent space. Due to strict point-wise minimization between predicted audio tokens and target embeddings, the adapter struggled to generalize across different musical styles. This resulted in a form of mode collapse, where predicted embeddings clustered into a nearly identical latent region, eliminating the visual diversity of the generated output. Consequently, we replaced the MSE objective with a InfoNCE Contrastive Loss and a Variance Loss. Through iterative empirical testing, we identified $\lambda_{\text{div}} = \lambda_{\text{var}} = 0.05$ as good weights, yielding a robust final training loss of 0.3426 and significantly more expressive visual results.

Stable Diffusion Hyperparameters

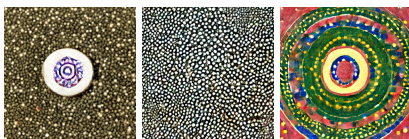
We conducted an empirical analysis of the guidance scale, which functions as a control mechanism to balance the model’s adherence to the conditioning signal against its generative stochasticity. Our experiments explored a wide range of values: lower scales (near 0) resulted in high creative variance but poor prompt fidelity, while higher scales (near 10.0) enforced strict prompt adherence. While standard text-to-image tasks typically utilize scales between 7.0 and 9.0, our results indicated that a lower value of 3.5 was better for our framework.



(a) Guidance = 0.2

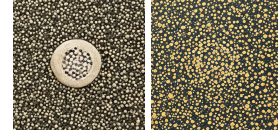


(b) Guidance = 3.5

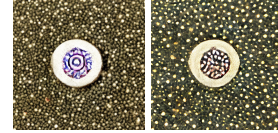


(c) Guidance = 10

While a guidance of 10 may initially yield images that appear qualitatively similar to those at 3.5, extended testing reveals a tendency toward semantic convergence, producing nearly identical visual frames despite significant variations in the input audio.



(a) Guidance = 3.5, white noise vs instrumental



(b) Guidance = 10, white noise vs instrumental

Additionally, we evaluated the impact of inference steps on the trade-off between image fidelity and system latency. Initial tests used 30 steps and produced sharp high-fidelity images; however, the generation speed was slow and therefore incompatible with the requirements of a real-time pipeline. By reducing the sampling process to 10 steps, we achieved a significant increase in generation speed while maintaining a sufficient level of visual quality and iconographic clarity.

The final adjustment was adding the negative prompt to better direct the output images.

B. Possible Improvements

Although the current version of the project operates as a rhythmic collage of images, future iterations could transition from a discrete collage to a continuous video stream by implementing temporal interpolation. Specifically, rather than treating each 8-second audio slice as an independent generation task, we could utilize Spherical Linear Interpolation (Slerp) to navigate the latent space between sequential audio-prompt embeddings. This would ensure a smooth semantic evolution of the imagery rather than an abrupt cut. Furthermore, integrating downstream frame-interpolation techniques, such as Optical Flow or AI-driven models like RIFE (Real-Time Intermediate Flow Estimation), would allow for the synthesis of intermediate frames, effectively eliminating visual instabilities and transforming the rhythmic montage into a fluid, cinematic audio-visual experience.

C. Application

A key outcome of this research is its upcoming application in a live concert setting this summer. We are partnering

with the Fabrica Harmonica A.P.S - E.T.S. association for a series of concerts themed around San Francesco. The project will be used to generate San Francesco themed images in real-time during the musical performances.

For the live shows, we intend to move away from purely random or single fixed seeds. Instead, we will select a set of seeds that produce high-quality results and run them in a circular rotation.

We have already conducted successful preliminary tests of this approach. The positive feedback from the music school has led to a formal collaboration to integrate this audio-reactive visual technology into their performances.